Assignment 9

Write a program that simulates the movement of a disk head based on a given sequence of cylinder requests and calculates the total head movement for each of the following disk scheduling algorithms:

1. SSTF (Shortest Seek Time First): Selects the request with the minimum seek time (closest cylinder) from the current head position.
2. SCAN (Elevator Algorithm): The disk arm moves from one end of the disk to the other, servicing requests in its path. When it reaches an end, it reverses direction and continues servicing requests.

Input:

Your program should accept the following inputs (e.g., via command line arguments or user prompts):

Initial Head Position: The starting cylinder number for the disk head.

Request Queue: A sequence/list of cylinder numbers representing pending disk requests.

Disk Size (Optional but Recommended): The maximum cylinder number (e.g., if cylinders are numbered 0 to 199, the size/max is 199). This is crucial for SCAN and C-SCAN. Assume cylinders are numbered starting from 0.

Initial Direction (for SCAN/C-SCAN): Specify the initial direction of head movement (e.g., towards higher cylinder numbers or lower cylinder numbers).

Output:

For each implemented algorithm, your program should output:

The name of the algorithm.

The sequence in which the cylinder requests were serviced.

The total amount of head movement (total seek distance in cylinders).